

Experiment 4

Jenkins installation and setup, explore the environment.

AIM: Download & Install Jenkins , login and explore the environment

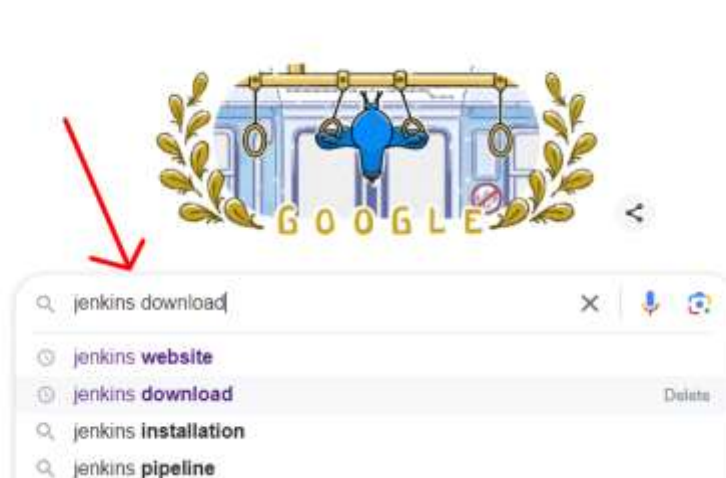
1.Detailed Guide to Install and Set Up Jenkins on Windows

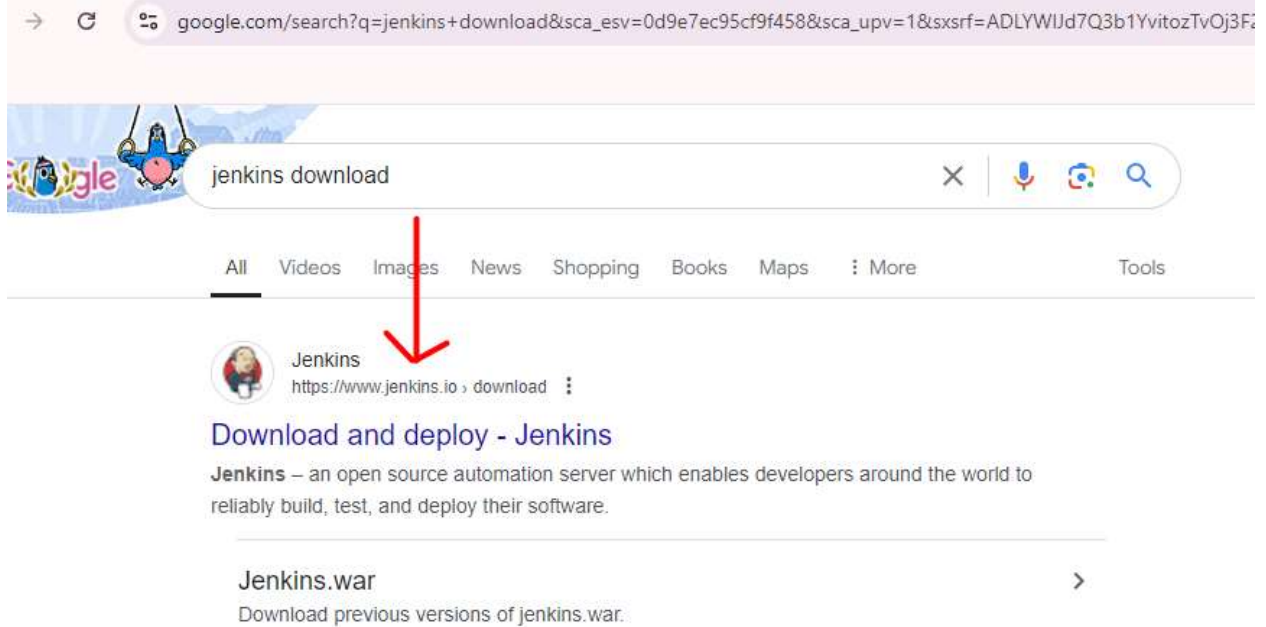
Prerequisites

- ❖ Ensure you have Java installed. Jenkins requires Java 8 or Java 11.
 - Download and install Java from the [Oracle website](#).
 - Set the JAVA_HOME environment variable:
 - Right-click on "This PC" or "My Computer" and select "Properties".
 - Click on "Advanced system settings".
 - Click on "Environment Variables".
 - Under "System variables", click "New" and add JAVA_HOME with the path to your Java installation (e.g., C:\Program Files\Java\jdk-11).

2. Download Jenkins

Visit Jenkins website





->Go to the Jenkins website (<https://www.jenkins.io/>)

->The Jenkins project produces two release lines: Stable (LTS) and weekly. We use Stable (LTS)

Downloading Jenkins

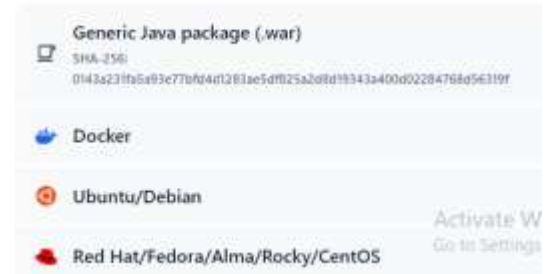
Jenkins is distributed as WAR files, native packages, installers, and Docker images. Follow these installation steps:

1. Before downloading, please take a moment to review the [Hardware and Software requirements](#) section of the User Handbook.
2. Select one of the packages below and follow the download instructions.
3. Once a Jenkins package has been downloaded, proceed to the [Installing Jenkins](#) section of the User Handbook.
4. You may also want to verify the package you downloaded. [Learn more about verifying Jenkins downloads.](#)

Download Jenkins 2.452.3 LTS for:



Download Jenkins 2.470 for:



- Click on Generic Java package
- jenkins.war file will downloaded and saved in downloads folder
- Once the jenkins.war is downloaded
- Open command prompt and go the downloads folder

```
C:\> Command Prompt

Microsoft Windows [Version 10.0.19045.4651]
(c) Microsoft Corporation. All rights reserved.

C:\Users\P c>cd Downloads

C:\Users\P c\Downloads>_
```

Type the following command

➤ **java -jar jenkins.war** and hit enter

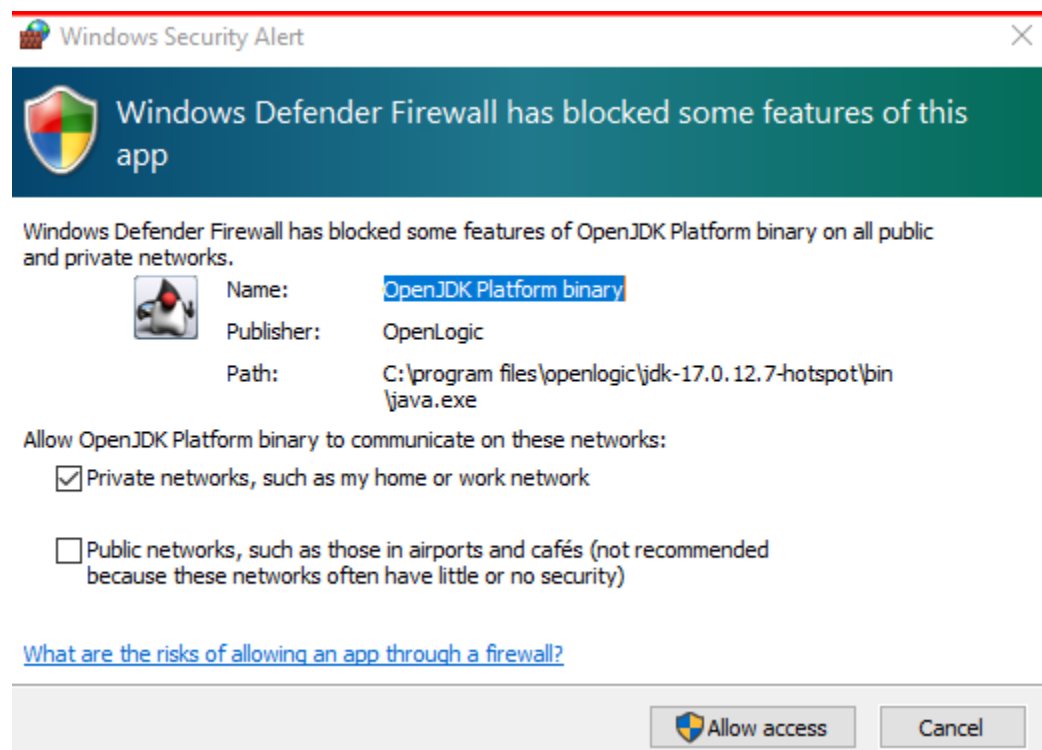
```
C:\> Command Prompt

Microsoft Windows [Version 10.0.19045.4651]
(c) Microsoft Corporation. All rights reserved.

C:\Users\P c>cd Downloads

C:\Users\P c\Downloads>java -jar jenkins.war
```

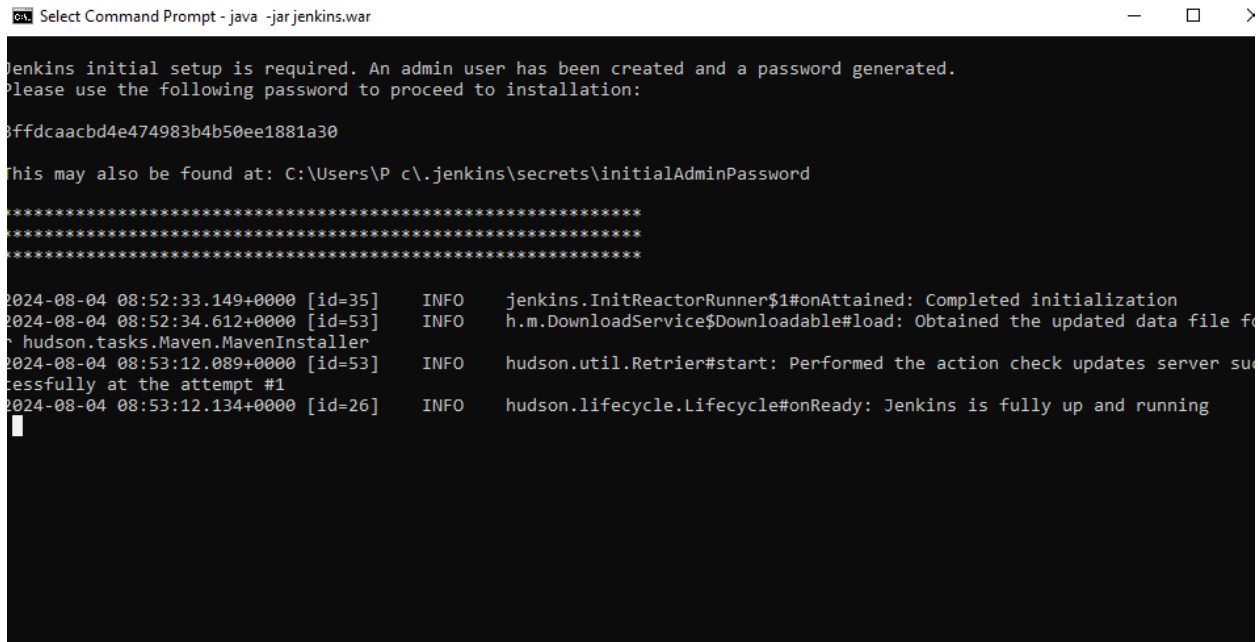
If you get below images then click on **Allow access**.



In the below image we get a message saying ***jenkins up and running***, it generates one password in the **C:\Users\P c\.jenkins\secrets\initialAdminPassword** file.

Minimize the command prompt.

Open the **initialPassword** file and copy the password.



```
Select Command Prompt - java -jar jenkins.war

Jenkins initial setup is required. An admin user has been created and a password generated.
Please use the following password to proceed to installation:

8ffdcacbd4e474983b4b50ee1881a30

This may also be found at: C:\Users\P c\.jenkins\secrets\initialAdminPassword

*****
*****
*****

2024-08-04 08:52:33.149+0000 [id=35] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
2024-08-04 08:52:34.612+0000 [id=53] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for
  hudson.tasks.Maven.MavenInstaller
2024-08-04 08:53:12.089+0000 [id=53] INFO HUDSON-UTIL-RETRIER#start: Performed the action check updates server suc
  cessfully at the attempt #1
2024-08-04 08:53:12.134+0000 [id=26] INFO HUDSON-LIFECYCLE-LIFECYCLE#onReady: Jenkins is fully up and running
```

Paste the password in the Administrator password textfield and hit continue.



In the below image select and press Install suggested plugins option.

Getting Started

Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

+

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.

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Activate
Go to Settings

Wait until all the plugins are installed.

Getting Started

Getting Started

☒ Folders	☒ OWASP Markup Formatter	☒ Build Timeout	☒ Credentials Binding	☒ OWASP Markup Formatter
☒ Timestampers	☒ Workspace Cleanup	☒ Ant	☒ Gradle	
☒ Pipeline	☒ GitHub Branch Source	☒ Pipeline: GitHub Groovy Libraries	☒ Pipeline Graph View	
☒ Git	☒ SSH Build Agents	☒ Matrix Authorization Strategy	☒ PAM Authentication	
☒ LDAP	☒ Email Extension	☒ Mailer	☒ Dark Theme	
** - required dependency				

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Create the Username , password and all necessary fields

K.Harshini
DevOps(cse)

Getting Started

Create First Admin User

Username

Password

Confirm password

Full name

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[Skip and continue as admin](#)

[Save and Continue](#)

By default we can see Jenkins URL as **localhost:8080** donot make any changes to it

Getting Started

Instance Configuration

Jenkins URL:

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the `BUILD_URL` environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

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[Not now](#)

[Save and Finish](#)

Save the Finish the process

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Getting Started

Jenkins is ready!

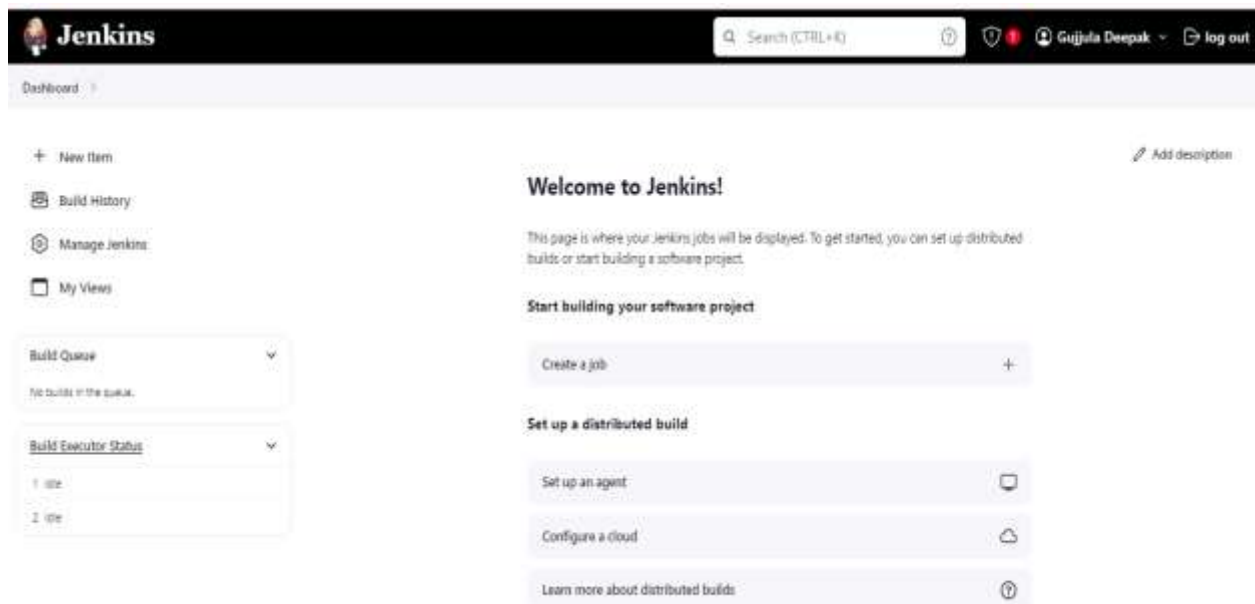
Your Jenkins setup is complete.

[Start using Jenkins](#)

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Jenkins are successfully installed. When you hit start using jenkins button you will get the below dashboard page

If not go to any browser and enter **localhost:8080** and hit enter



The screenshot shows the Jenkins dashboard interface. At the top, there's a black header with the Jenkins logo, a search bar (Search (CTRL+K)), and user information (Gujjala Deepak) with a log out button. Below the header, the main content area is divided into two columns. The left column contains a sidebar with links: New Item, Build History, Manage Jenkins, and My Views. Below these links are two expandable sections: 'Build Queue' (showing 'No builds in the queue') and 'Build Executor Status' (showing '1 idle' and '1 idle' executors). The right column features a 'Welcome to Jenkins!' message, followed by a paragraph explaining the dashboard's purpose. Below this, there's a section titled 'Start building your software project' with a 'Create a job' button. Further down, a 'Set up a distributed build' section includes links for 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'.

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Exploring the Jenkins Environment in Detail

In the previous image we can see all necessary things to create a project

Dashboard

New Item:

Create new projects in Jenkins

Types of Projects:

1. **Freestyle Project:** A flexible and easy-to-setup project type. You can define simple jobs and chains of jobs.
2. **Pipeline:** Allows you to define your build process as code. You write a script in Groovy language to automate your workflow.
3. **Multi-configuration Project:** Used for running the same job with different configurations (e.g., testing on multiple platforms).
4. **Folder:** Organize multiple jobs into a single folder.
5. **GitHub Organization:** Automatically create Pipeline jobs for each repository in a GitHub organization.
6. **Multibranch Pipeline:** Automatically create a pipeline for each branch in a repository.

Example:

- To create a new Freestyle project:
 1. Click on "New Item".
 2. Enter the name of the project.
 3. Select "Freestyle project" and click "OK".
 4. Configure the project (e.g., source code management, build triggers, build steps).
 5. Save the project.

Build History:

Track and display the status of recent builds.

Information Available:

- Build number.
- Project name.
- Build status (success, failure, unstable).
- Timestamp of the build.

Example:

On the main dashboard, you can see a list of recent builds under "Build History". Clicking on any build number will provide detailed information about that build.

Manage Jenkins:

Access to administration and configuration options for Jenkins.

Options Available:

- Configure System: Set global configurations.
- Global Tool Configuration: Manage tools (e.g., JDK, Git, Maven).
- Manage Plugins: Install, update, or remove plugins.
- Security: Configure global security settings.
- System Information: View detailed information about the Jenkins environment.
- Manage Nodes: Manage slave nodes for distributed builds.
- Script Console: Run Groovy scripts for advanced configuration and automation.

Example:

Navigate to "Manage Jenkins" to see all the available administration options.

Result: Verified that I was able to install and log in to Jenkins, and explored the environment